Rotation speed transmiter BR4



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1. Purpose and properties

Speed transmitter is used for rotation speed measurement of both asynchronous and synchronous generators directly on output terminals (without any special sensor). The transmitter is able to measure the speed of a disconnected generator too, due to the low AC voltage generated by the generator proportionally to the speed due to the remanent magnetism. The transmitter is able to measure the frequency of this low voltage as well as the frequency of connected generator voltage which is defined by the electrical network.

The transmitter offers the analogue output (4-20mA) proportional to the frequency. This output can be used as an analogue input for control system. Galvanic isolated pulse output can be used for frequency input of control system. The frequency of this pulse output is doubled compared to the generator frequency.

The transmitter offers two contact outputs too. The first can be used for synchronizing of an asynchronous generator, the second one can be used as an overspeed protection.

2. Description

Measured AC signal (1 - 450V) is in galvanic isolated part converted to pulses of doubled frequency. These pulses are used directly as the pulse output signal and simultaneously as an input to microprocessor part where they are measured with high accuracy each 10ms.The microprocessor is controlling DA converter which send 4-20mA analogue output.

Microprocessor also control two output relays . The first relay output (**Sync**) is closed when adjusted synchronizing frequency is reached (green LED is ON). The second one (**Oversp**) is open on contrary when adjusted overspeed frequency limit is reached (red LED is ON).

Measured frequency is approximately indicated by means of the ten segments bar.

Pulse output LED is red blinking during the operation.

3. Functions

Unified output pulses of doubled frequency. Pulse level is 20V, the length is approximately 2ms.

Frequency converter to analogue signal 4-20mA.

Ten segments bar indicator of the measured frequency.

Output relay (closing) when frequency reached adjustable value (47 – 50 Hz).

Output relay (disconnecting) when frequency reached adjustable value (65 – 80 Hz)

4. Setup

Two adjusting trimers are accessible from the front side. The first one for „Sync“ level setting (47–50 Hz), the second one for „Overspeed“ level setting (65-80 Hz).

5. Technical data

Dimensions (HxWxD)	75x70x110mm
Power supply	24VDC
Input Power	< 3W
Internal fuse	not used
Measured signal	1 – 450VAC, shortly 550VAC (two possible inputs 230V~ or 400V~)
Input galvanic isolation	yes
Output pulses	0 +20V, 2ms, max. 2mA, 0-200Hz
Analogue output	active 4–20mA = 0–100Hz
Output relays	5A, 230VAC, 30VDC
Installation	DIN rail

6. Wiring diagram

